Reg. No. [ ][ ][ ][ ][ ][ ][ ][ ][ ][ ] **[18AN71]**

SEVENTH SEMESTER B.Tech EXAMINATION, DECEMBER–2021

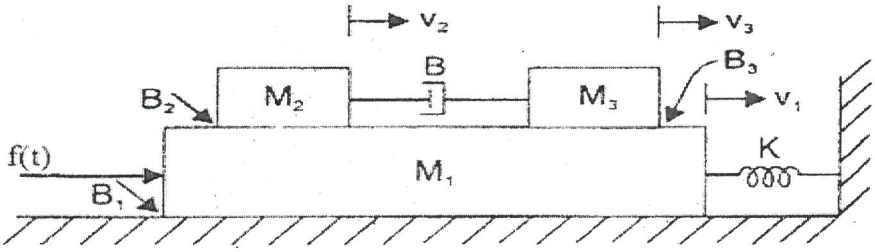
CONTROL SYSTEMS

Time: 3 Hours **Maximum Marks: 70**

Instructions: i. Answer **FIVE** full Questions. Question number 1 and 2 are compulsory. Each question carries 14 marks.

ii. Missing data may be suitably assumed.

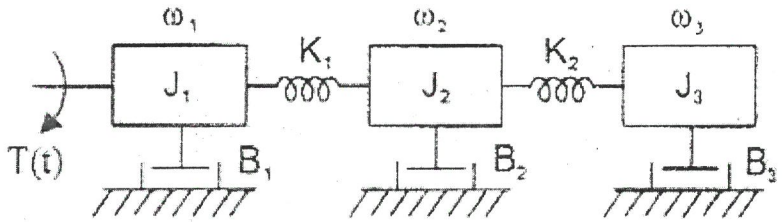
1. a. Write the differential equations governing the mechanical system shown below. Also draw the **i.** force voltage **ii.** force current analogous circuits **07**



b. Write the differential equations governing the rotational mechanical system shown below. Also draw the **07**

i. torque voltage

ii. torque current analogous circuits



2. a. Using Routh Hurwitz Criterion determine the location of the roots of the characteristic equation and comment on the stability of the system. **07**

$s^7 + 9s^6 + 24s^5 + 24s^4 + 24s^3 + 24s^2 + 23s + 15 = 0$

b. A unity feedback system has the forward path transfer function $G(s) = \frac{10}{(s+1)}$. Find the steady state error and the generalized error coefficient for r(t) = t. **07**

3. a. Sketch the root locus for a unity feedback system that has the forward transfer function **07**

$$G(s) = \frac{K}{(s+2)(s+4)(s+6)}$$

b. Sketch the bode plot for the transfer function **07**

$$G(s) = \frac{(s+3)}{(s+2)(s^2+2s+25)}$$

OR

4. a. Sketch the polar plot for the transfer function and find the gain and phase margin **07**

$$G(s) = \frac{K}{s(1+s)(1+2s)}$$

b. Explain in detail about the PID controller with a neat diagram **07**

5. a. What is a compensator? What are its types and describe in brief about the same. **07**

b. Explain in detail about the lead compensator. **07**

OR

6. a. Explain in detail about the lag compensator. **07**

b. Explain in detail about the lead-lag compensator. **07**

7. a. Determine whether the system **07**

$$\dot{\mathbf{z}} = \mathbf{A}_z\mathbf{z} + \mathbf{B}_z u = \begin{bmatrix} -5 & 1 & 0 \\ 0 & -2 & 1 \\ 0 & 0 & -1 \end{bmatrix} \mathbf{z} + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} u$$

is controllable or not.

b. Determine whether the system **07**

$$\dot{\hat{\mathbf{z}}} = \mathbf{A}\hat{\mathbf{z}} + \mathbf{B}u = \begin{bmatrix} -7 & 1 & 0 \\ 0 & -8 & 1 \\ 0 & 0 & -9 \end{bmatrix} \hat{\mathbf{z}} + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} u$$

is observable or not.

OR

8. a. Derive the expression for observable canonical form. **07**

b. Derive the expression for diagonal canonical form. **07**